AR-Mini · System Architecture v1

Controller v1: one XIAO ESP32-S3 Sense runs the whole robot · power is fused per branch, signals only pass through the XIAO and the PCA9685 · 06.10.2026

Power



Turnigy 3S 2200 mAh
11.1 V · 40 C · XT60

12 V



Inline fuse 30 A
GT4652 · on the XT60 lead

12 V



Main switch 50 A
R13-401A2A · cuts everything

12 V



6-slot fuse block
branches below

Fused branches (12 V)

- F1 20 A — ESC + thrusters ×2
- F2 15 A — servo bus driver
- F3 10 A — BTS7960 ×2 → tracks
- F4 5 A — logic buck → 5 V
- F5 5 A — Pi buck (v2)



F4 5 A — logic rail

MZHOU buck
12 → 5 V · 3 A



F5 5 A — Pi 5 rail

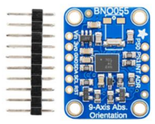
QIQIAZI buck
12 → 5 V · 5 A

Charged off-robot on SkyRC iMAX B6 V2

Sensing



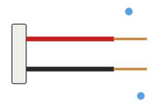
MLX90642-D45
thermal · forward 45°



BNO055 IMU
9-DOF · heading / roll



MLX90640 110°
thermal · on the cradle



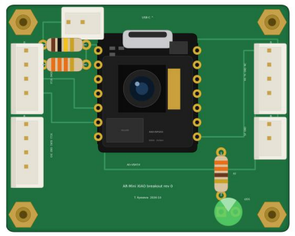
Leak probe
2 bare wires · hull floor



VBAT divider
100 k / 33 k → ÷ 4

I²C-0 (D4 · BNO055 · PCA9685)

Control



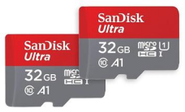
XIAO ESP32-S3 Sense
on breakout rev 0 (own PCB, 50 × 40 mm)

on board

A0 VBAT · A1 LEAK + KILL · D2 / D3 I²C-1 · D4 / D5 I²C-0 · 5 V in · D6 / D7 UART1 · USB-C → Pi (v2) · 5 V logic



OV3660 camera
on XIAO Sense · arch pan



microSD log
CSV 10 Hz + snapshots

Firmware (FreeRTOS)
core 0 perception · core 1 control

Mission state machine (core 1)

SEARCH → APPROACH → ALIGN → LIFT → RETURN

any state → SAFE on KILL · LEAK · LOW BATTERY · lost target

XIAO 3V3 → sensors + PCA9685 VCC · 4.7 k pull-ups · common GND everywhere

v2 · 2027 (not in v1)



Raspberry Pi 5 · 8 GB
ROS 2 Jazzy · rosbag



Active cooler
fan + heatsink

next:
replaces the XIAO

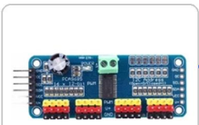


ESP32-S3-WROOM-1
carrier rev A · N16R8



ESP32-S3-DevKit-Lipo
firmware test board

Actuation



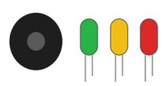
PCA9685
16-ch PWM · I²C 0x40

CH0 / CH1 · OE



Logic kill (rocker)
ON-ON 20 A

CH6–9



Buzzer + 3 LEDs
CH6 · CH7–9

CH2–5



Thruster L + ESC
Cuifati 560 KV · CCW



Thruster R + ESC
Cuifati 560 KV · CW



BTS7960 L
43 A H-bridge



BTS7960 R
43 A H-bridge

M+ / M−



DFRobot ROB0154 tracks
2 DC motors · axis 62 mm above the hull floor

UART1



Serial bus servo driver
Waveshare · 1 Mbps

ID 1–4 on one 5264 daisy-chain (half-duplex, 1 Mbps)



ST3215 · ID 1
arm left



ST3215 · ID 2
arm right



ST3215 · ID 3
camera pan



ST3215 · ID 4
spare / v2

30 kg·cm @ 12 V · magnetic encoder · position / load / temperature feedback

Parallelogram lift: arm 110 mm · deploy 215° / stow 45° · cradle Ø5 rod 220 × 123 mm

Thrusters: differential steering on water · tracks: ROB0154 on land, motor axis above the waterline

Wires: 12 V (fused branch) · 5 V · 3V3 · I²C-0 · I²C-1 · UART / servo bus · PWM · analog / kill · motor power · USB (v2, dashed)

F1 20 A = fed from that fuse-block slot (12 V). Common GND for every module. Kill: 50 A main switch (hardware) + logic rocker → PCA9685 OE + ADC ladder on A1.